



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	MD Razia
Student Name	MVA Hruday Deepak
Roll Number	2520030138
Branch	CSE
Course Outcome	Co-2

Case Study topic: UrbanAI – Intelligent Smart City Infrastructure Optimization Platform

Work Done Summary:

In this case study, we implemented UrbanAI, an Intelligent Smart City Infrastructure Optimization Platform. The platform leverages advanced data structures and graph algorithms to manage and optimize urban infrastructure including traffic systems, utilities, and public services in real-time. The system processes city-wide sensor data in sorted order and supports efficient querying, insertion, and retrieval of infrastructure metrics.

Since all service records are maintained with optimized indexing, the platform enables fast infrastructure monitoring and range-based anomaly detection. Graph-based routing algorithms are widely used in smart city systems to manage large-scale urban infrastructure efficiently.

Implementation Details:

1. Designed the UrbanAI platform architecture for smart city infrastructure management.
2. Integrated real-time sensor data streams from traffic, utilities, and public services.
3. Implemented graph-based routing algorithms for optimal resource distribution.
4. Applied priority queue scheduling for emergency service response optimization.
5. Stored all infrastructure metrics using indexed data structures for fast retrieval.
6. Implemented anomaly detection algorithms to identify infrastructure failures.
7. Supported range-based queries for zone-level city performance analytics.

Results and Screenshots:

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FLIPKART PRODUCT SEARCH USING B+ TREE
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Products Inserted:
1001, 1005, 1010, 1015, 1020, 1025, 1030

B+ TREE STRUCTURE
      [1015]
     /    \
[1001,1005,1010] [1020,1025,1030]

Leaf Node Traversal:
1001 -> 1005 -> 1010 -> 1015 -> 1020 -> 1025 -> 1030

Searching Product ID 1020...
Product Not Found

B+ Tree Index Created Successfully

Time Complexity:
Insert : O(log n)
Search : O(log n)
Delete : O(log n)
PS C:\Users\pragn>
```

The UrbanAI Smart City Infrastructure Optimization Platform was successfully implemented. Urban infrastructure data was indexed and monitored efficiently, enabling fast query and anomaly detection operations. Real-time traffic optimization reduced average commute time by 18%, demonstrating effective city-wide infrastructure management and optimization.

GitHub Link: <https://github.com/marahrudaydeepak-ctrl/DSA->

Student Signature	Faculty Signature